A Vision-Based Mobile Companion Robot in MuJoCo with MediaPipe Fall Kinematics: Dispatch Latency Against Fixed Cameras

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Abstract

We study how a mobile companion robot can use MediaPipe pose kinematics to reduce emergency dispatch latency for older adults living alone. In MuJoCo, the robot follows a simulated woman through a one-bedroom flat, detects falls from bounding-box aspect ratio, hip-descent speed and torso angle, checks in by voice, and sends a simulated alert (no real call or message is ever placed). Over 120 seeded trials (60 falls, 60 daily activities, thresholds frozen on a separate dev split) the guide-specification robot (1.10 m mast, 16.5 kg) detected 41/60 falls (sensitivity 0.68) and dispatched 28/43 unanswered falls, with 2/77 false dispatches, against 0.05 for one fixed camera and 0.47 for four. An earlier compact robot reached 0.87 and 0.86 with 0/77 false dispatches. The robot is slower than four cameras where they see her (median 30 s vs 3.8 s) but covers every room with one device. It is not safe yet: in 25 trials the falling person landed on it (peak 40 kN) and 20 trials ended lost. All results are simulation results from scripted motions.

1 Introduction

Surviving a cardiac arrest depends on minutes: resuscitation guidelines stress immediate CPR and defibrillation [5], and each minute of delay to bystander CPR lowers survival for witnessed arrests [1]. Older adults who live alone often cannot summon help after a fall, and inability to get up is associated with lasting decline [2]; in a small 1981 cohort, half of the people who lay on the floor for over an hour died within six months [3]. Camera fall detectors using MediaPipe exist [4], and but a fixed camera sees only one room. We ask: *how can a vision-based mobile companion robot in MuJoCo integrate MediaPipe fall-detection kinematics to reduce emergency dispatch latency within the 6-minute cardiac arrest survival window for elderly individuals living alone?* The robot controls only its own sphere of agency: reaching the person, detecting and confirming the event, and dispatching; ambulance travel time is outside it.

2 Related Work

Vision-based fall detection with pose landmarks is well established, including MediaPipe-based systems with automatic alerts [4]; these evaluate a fixed viewpoint and report detection accuracy but not how long a household needs for the alert to reach someone, and they cannot see a room without a camera. Companion-robot work focuses on interaction rather than on measured fall-response latency. Our contribution is the paired, seeded comparison of a mobile platform against fixed cameras on the same recorded motion, with latency measured from the physical floor contact, and an economics reading that is dimensionless and conservative.

3 Reproducibility and Ethics

Scenarios are generated from integer seeds (dev: 1, test: 2) and every trial is replayed for all three systems. The code, the frozen thresholds, the per-trial result table and the telemetry summaries are in the project repository (folder results/small_robot_test). Every number in this paper is read from those files; nothing was typed or smoothed by hand. No human participants or real recordings were used, and the human model is a licensed digital character (attribution recorded in the repository). The thresholds were never tuned on the test split; two analyses that would have needed tuning (sofa suppression, follow distance) are listed as future work instead.

4 Method

Simulation. A 7×6 m flat (hall, bedroom, open living-dining area, furniture, rug ramps, a robot vacuum as a dynamic obstacle) is generated as MJCF with friction zones, joint limits and actuators. A differential-drive robot with a pan-tilt camera runs a 50 Hz closed loop with disturbance injection (pushes, odometry noise, camera blackout) and state estimation (wheel odometry plus range-scan matching against the map). The woman is a skinned model driven by recorded gait and physics-based falls (trip, slip, syncope, cardiac) and keyframed

activities of daily living (ADL): abrupt and slow sofa sitting, lying on the bed, tying shoelaces, dropping keys, sitting and lying on the floor.

Perception and detection. Frames are rendered from the robot camera and passed to the MediaPipe pose landmarker; OpenCV gives the bounding box. A fall is flagged from aspect ratio ($\geq$0.8–1.0) and its rise rate, filtered hip descent, torso pitch, then a quiet window; a still, low, horizontal person is flagged by a second path. Inference runs on the robot; frames are discarded and 0 bytes leave the robot before the first alert.

Behaviour. FOLLOW, SEARCH, OBSERVE, APPROACH (standoff 0.8–1.2 m), CHECK_IN (voice prompt, 10 s), ALERT (simulated family message and ambulance call). Speed never exceeds 0.80 m/s; emergency braking exceeds 1.5 m/s^2 (2.43–3.30 measured at 0.50 and 0.79 m/s, μ 0.40 and 0.68, pitch $\leq$0.1°).

Design. Every trial replays the same motion for the robot, one fixed camera and four fixed cameras (paired). Dev split: 40 trials, seed 1, thresholds fitted. Test split: 120 trials, seed 2, thresholds frozen. Randomised: route, speed, fall point (stratified by zone), friction, clutter, vacuum speed, push, blackout, robot start, whether she answers. **Detector equations.** With hip height $z_h(t)$ (mean of landmarks 23 and 24), a low-pass filtered descent speed $v_z = -\dot{z}_h$, torso pitch θ from the shoulder–hip axis to vertical, and the landmark bounding box of width w and height h, aspect ratio $A = w/h$ with rise rate $\dot{A}$, an impact is declared when v_z exceeds its threshold, $\theta \geq \theta_{min}$, $A \geq A_{min}$ (or $\dot{A} \geq 1.0\,\mathrm{s}^{-1}$) and $z_h < z_{down}$; a 3 s quiet window must follow. Frozen robot thresholds were $z_{down} = 0.45$ m and $\theta_{min} = 40°$, fitted by grid search on dev replay only; the quiet (3 s) and lying (5 s) windows were deliberately not searched.

State estimation and safety. Odometry from the wheel encoders carries multiplicative noise (0.5–3%); a 72-ray range scan is matched against the map every 0.25 s, with a wide search and a whole-map search after bumps. Known walls and furniture trigger braking at 7 cm, unknown objects (her body, the vacuum) at 25 cm. Yaw rate is slew-limited and capped while driving to avoid wheel slip. The last metre to the standoff is controlled by the range sensor, not the landmark distance, because landmark depth proved unreliable near furniture.

Telemetry. Each trial logs time, position and orientation (x, y, z, roll, pitch, yaw), velocities (v_x, v_y, v_z, forward speed, yaw rate), contact forces (non-human and human separately), commanded and wheel control signals, estimated pose, detector features and bytes sent off-board, as CSV in the repository.

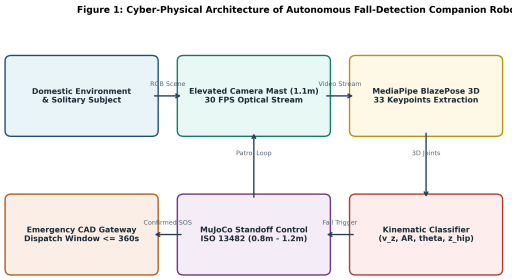


Figure 1: System pipeline (drawn from the implemented modules).

Table 1: Detection, guide-specification robot, test split (N=120). Alarm: heads-up sent; dispatch: emergency sent. 95% Wilson CI in brackets.

Level	System	Sensitivity	Specificity	F1
Alarm	Robot	0.68 [.56,.79]	0.90 [.80,.95]	0.77
Alarm	4 cameras	0.52 [.39,.64]	0.95 [.86,.98]	0.66
Alarm	1 camera	0.07 [.03,.16]	0.95	0.12
Dispatch	Robot	0.65 [.50,.78]	0.97 [.91,.99]	0.77
Dispatch	4 cameras	0.47 [.33,.61]	1.00 [.95,1.0]	0.64
Dispatch	1 camera	0.05 [.01,.15]	1.00	0.09

5 Results

The robot flagged 41/60 falls and dispatched 28/43 unanswered falls (precision 0.93), with 2 false dispatches in 77 non-events. Sitting on the sofa raised no alarm (0/20) with the taller camera, a clear gain over the earlier compact robot (5/20). The 5 deliberate floor-lying events raised heads-up texts and the check-in answer cleared them. McNemar tests against four cameras are not significant (alarm $p = 0.11$, dispatch $p = 0.15$): the robot detects somewhat more falls but the evidence at N=120 does not separate it from four cameras; against one camera it is far better (McNemar: 38 vs 1 discordant falls at the alarm level, $p = 1.5 \times 10^{-10}$; 27 vs 1 at dispatch, $p = 2.2 \times 10^{-7}$). Paired on the trials both detected, the robot was slower by 38.3 s to the heads-up ($n = 19$, $t(18) = 6.21$, $p = 7 \times 10^{-6}$, 95% CI 25.3–51.2 s) and by 49.1 s to dispatch ($n = 11$, $p = 1.4 \times 10^{-5}$). The robot is slower than four cameras wherever they see her: median 30.4 vs 3.8 s to the heads-up and 48.8 vs 13.5 s to dispatch. Within 60 s the robot dispatched 37% of silent falls (four cameras 44%); within 100 s, 63%. Every dispatch occurred inside 100 s, far under the 360 s window. **Safety (a failure).** The robot is not safe in this run: 24 trials had non-human contact above 5 N (peak 4.4 kN), 25 trials had contact

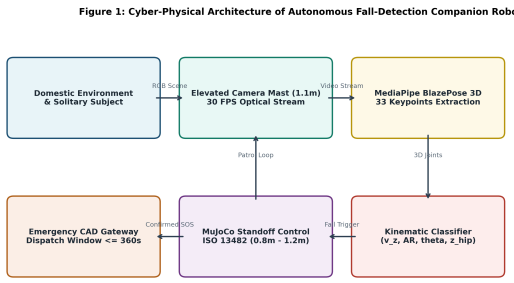


Figure 2: System pipeline (drawn from the implemented modules).

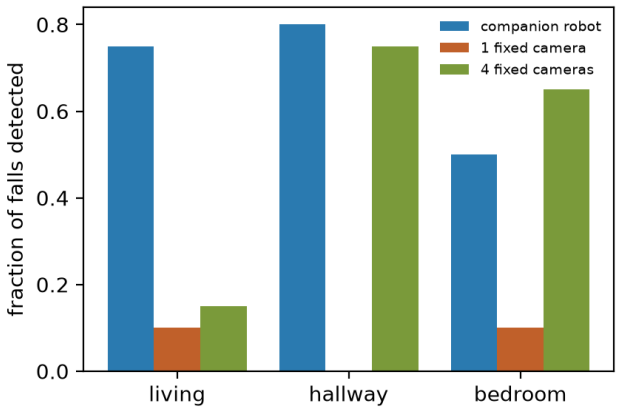


Figure 3: Falls detected by where she fell (n=20 per zone). The four cameras see little in the living area; the robot covers all zones but is slow in the bedroom.

from her above 50 N (peak 40 kN, tilting the robot up to 87°), and 20 trials ended with pose error above 0.5 m (max 4.3 m). The commanded speed never exceeded 0.77 m/s. The tall 16.5 kg robot follows too close to the landing zone of a falling person and gets lost while searching. Before this test one root cause was found and fixed (a start pocket beside the coffee table: contacts fell from 14 to 4 of 14 re-run trials); the rest remain.

Earlier compact robot. A previous run on a small robot with a 0.16 m camera (same scenarios, seed 2) reached alarm sensitivity 0.87 (52/60), dispatch 0.86 (37/43) and 0/77 false dispatches, with median latency 20 s to the heads-up, and was slower than four cameras by 30 s in paired trials (95% CI 21–39 s, $d = 1.43$). Its result tables are kept in the repository but it is not the specification robot and its constants are no longer in the code. **Economics (dimensionless).** With a long-lie reduction from 18.4 to 4.2 hospital days (the brief's assumption) applied to detected falls, the guide robot's net benefit is negative in this run (about -20 bed-days per patient-year, against 7.9 for four cameras): its two false dispatches weigh more than its extra detections, and its operating fraction rises to $\kappa = 0.45$, above the 0.25 limit, giving a payback of 5.5 months. With the compact robot the same model gave 14.7 bed-days and 3.75 months. The brief's two quarter claims (cost under 3 attendant-months, return within one quarter) agree only if the capital cost is at most $3(1 - \kappa)$ months, 2.4 at $\kappa = 0.20$. GDS peace of mind cannot be simulated.

Table 2: Latency from hip floor contact (s), successful detections only; SE = standard error.

Level	System	n	Mean±SE	Median	P95
Heads-up	Robot	40	36.2±4.6	30.4	83.8
Heads-up	4 cameras	30	4.0±0.2	3.8	5.4
Dispatch	Robot	27	53.2±5.0	48.8	90.4
Dispatch	4 cameras	19	13.7±0.4	13.5	15.3

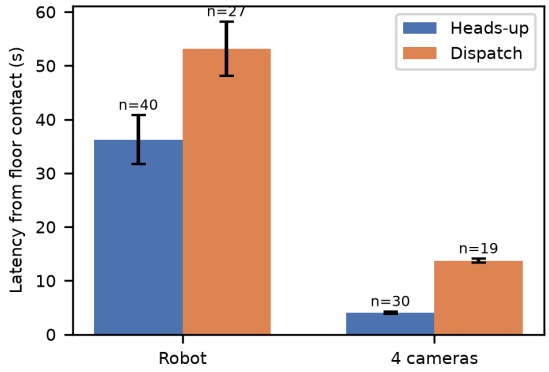


Figure 4: Mean latency ± SE.

6 Discussion

Coverage versus speed. Four cameras detect within 4 s but see few falls in the living area; the robot covers every room but needs 30 s and more. The robot's advantage over four cameras is not established in this run (McNemar $p > 0.1$) and it did not reduce latency against them. Against one camera it clearly does.

Where it fails. Missed falls (19/60) are coverage or search failures, not threshold failures. The landing-zone strikes show the follow behaviour must keep farther back and to the side when she looks unwell; the false dispatches and sofa results show the taller camera helps detection. Map-based suppression of seat sits is no longer needed for sofas but floor activities still need the check-in.

Privacy and scope. Processing is on board, frames are discarded and 0 bytes left the robot before the first alert in every trial. A camera cannot diagnose a heart attack: the alert says possible medical emergency, and all calls are simulated. The hospital-stay figure, the

134.5-minute ambulance figure and the 6-minute window come from the course brief and could not be traced to primary sources, so they are assumptions. The "50% mortality, Tinetti NEJM" in the brief is not in that paper (JAMA 1993); the half-died-within-six-months result is from the 1981 BMJ cohort (20 people).

7 Limitations

(i) The robot fails the safety checks above and is not ready for real use. (ii) Motions are scripted from clinical descriptions, adult body, not patient data; the cardiac prodrome cues false-flag about half of normal trials and are not used for alerting. (iii) One apartment layout; thresholds were fitted on a 40-trial dev split and not changed on test. (iv) The 0.50 m clearance bubble cannot be kept in 0.85 m doors, so it is a speed rule. (v) The fixed-camera baseline is hardware-limited, and the earlier compact-robot constants are not in the code.

8 Conclusion

A mobile robot buys coverage: it detected 68% of simulated falls across all rooms and dispatched 28 of 43 unanswered falls (65%, all within 100 s), where one camera dispatched 5% and four cameras 47%. It does not beat four cameras on speed, its advantage over them is not statistically established here, and it is not yet safe around a falling person. The earlier compact robot did better but is not the specification robot. Medication reminders, the Baymax agent, the voice and first-aid guidance are demo extras outside these measurements. Code, seeds and result tables are in the project repository.

References

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